Pranshu Tripathi

Software Engineer

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SUMMARY

Software engineer at Schrödinger building GenAI platforms across Maestro Assistant and Bunsen, spanning multi-agent workflows, user-controlled agent automation, and production reliability. Delivered a 300–400x C++ molecular-calculation speedup and eliminated a blocking ~60-second Bunsen startup experience.

EXPERIENCE

Schrödinger

Senior Developer I *January 2026 - Present*

- Led Maestro Assistant, a GenAI platform, shipping its multi-agent backend and durable session and conversation persistence with configurable knowledge-bot behavior.
- Closed a critical Maestro Assistant trust-boundary gap and delivered LLM-driven failed-job debugging that turns raw logs into actionable explanations.
- Built Bunsen's opt-in Auto mode and approval controls, allowing long-running agent jobs to proceed without repeated prompts while retaining explicit user control.
- Shipped an owner-only LLM resiliency dashboard that made outages diagnosable and eliminated a blocking ~60-second Bunsen pod-startup experience.
- Hardened Bunsen against proprietary-asset exfiltration and stale sessions, and restored stable request latency by resolving production database connection leaks.

Developer II *July 2024 - December 2025*

- Built a modular C++ molecular-interaction library with caching, reducing calculation time 300–400x and cutting 10-minute tasks to 30 seconds.

Developer I *June 2023 - June 2024*

- Refactored 2000+ lines of legacy code using Factory and Strategy patterns, improving extensibility for handling diverse Job Server outputs.
- Enhanced export functionality for new movie presets (e.g., View Ligand Docking) from Maestro to PyMOL, decreasing manual visualization setup time by 60% and enhancing automated workflow efficiency.

Intern - Software Development *January 2023 - May 2023*

- Created an adapter for Schrödinger's structure class that removed the m2io dependency and enabled a property-writing mechanism with 3x greater flexibility, gaining recognition from Scientific Infrastructure team.
- Implemented a robust Find Toolbar widget following MVC pattern and command pattern with python and C++ inter communication.

Intern - Software Development *May 2022 - July 2022*

- Delivered a robust preset management feature in 2 months—30% faster thanaverage—using Factory and Adapter design patterns.

PROJECTS

EventHive

Deployed a ticketing platform on GKE handling 3,000+ RPS, publishing NPM packages to streamline microservice integration across 5 services

EDUCATION

Georgia Institute of Technology

M.S. in Computer Science
August 2026 - Present

- Coursework: High Performance Computing.

IIIT Allahabad (B.Tech)

Electronics and Communication
Aug 2019 - May 2023

- **GPA: 9.16/10**
- **Batch Rank: 4**
- **Award:** Merit-cum-Means Scholarship for Academic Excellence 2020, 2021, 2022

AWARDS

- Appreciation from CEO, CTO for Maestro Assistant Project

SKILLS

Programming Languages
- C++, Python, TypeScript, SQL

Framework & Libraries
- FastAPI, React, Qt, SQLAlchemy, Alembic, asyncio, Auth0, OIDC, Server-Sent Events

Cloud & DevOps
- Docker, Kubernetes, GCP, GCP Logging, Observability

Generative AI Technologies
- LLM, LLM Agents, Tool Calling, RAG

Additional Interests
- Production Systems, LLM Applications

LANGUAGES

English, Hindi, Marathi, German